MIL-P-16685C
12 October 1962
SUPERSEDING
MIL-P-16685B
17 February 1961
JAN-P-534
2 December 1947

MILITARY SPECIFICATION

PACKING, ROUNDED, BULK AND PREFORMED

(STUFFING-TUBE FOR ELECTRIC CABLES)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. - This specification covers rounded packing on spools or reels and preformed packing for electric cable seals in stuffing tubes.

1.2 Classification. - Packing shall be of the following classes, parts, and types as specified (see table I, figures 1 and 2 and 6.2):

Class 1 - Rounded packing on spools or reels.

Part A - Retainer packing.
Part B - Sealing packing.

Class 2 - Preformed.

Type I
Type IA
Type II

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids, forms a part of this specification to the extent specified herein:

STANDARD

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N.Y.)

FSC 5330

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3. REQUIREMENTS

3.1 Preproduction. - Prior to production, a preproduction item shall be required (see 4.2).

3.2 Class 1, rounded. -

3.2.1 Material. -

3.2.1.1 Part A. - Packing shall be composed of asbestos rovings or yarns, either braid over braid or braided square, and treated with a waterproof compound.

3.2.1.2 Part B. - Packing shall be composed of an admixture of cork particles, graphite, lubricant and a plastic binder enclosed in a woven cotton lattice jacket.

3.2.2 Sizes. - Packing shall be furnished in rounded cross section in sizes 1/8 to 1/2 inch, inclusive, in increments of 1/16 inch, as specified (see 6.2).

3.3 Class 2, preformed. -

3.3.1 Material. -

3.3.1.1 Piece 1. - Piece 1 shall be made of packing conforming to class 1, part A, as shown on figures 1, 2 and 3.

3.3.1.2 Piece 2. - Piece 2 shall be made of packing conforming to class 1, part B, which may be made in one continuous piece in the form of a tight helix, or may be cut into individual rings (see figures 1, 2 and 4). If made of individual rings, the cuts shall be displaced approximately 90 degrees from the cuts in the adjoining rings. The cut in each top and bottom retainer rings shall be similarly displaced from the cut in the adjoining sealing component.

3.3.2 Coating. - After molding, packing shall receive a thin coat of a suitable quick-drying resin preparation. The coating shall be of such a nature that it will in no way affect the sealing properties of the class 1, part B packing when tested as specified in 4.5.

3.3.3 Sizes. - Packing shall be of the sizes shown in table I and as specified (see 6.2).

Table I - Sizes.

Size	d 1/ Inch	Dimensions				
		outside diameter Inches	inside diameter Inches	h-1 (type I) Inches	h-2 (type II) Inches	h-1A (type IA) Inches
A	31/6	0.750	0.375			
B	3/16	.875	.480			
C	3/16	1.000	.610			
D	3/16	1.125	.719	1-3/8 ± 1/16	1-1/8 ± 1/16	Same as h-1
E	3/16	1.125	.778			
F	3/16	1.187	.812			
G	1/4	1.375	.911			
J	1/4	1.500	1.037			
K	1/4	1.625	1.120			
L	1/4	1.687	1.232			
M	1/4	1.875	1.375			
N	1/4	1.937	1.474	2-5/8 ± 3/32	1-3/8 ± 1/16	1-11/16 ± 1/16
P	1/4	2.062	1.586			
R	5/16	2.187	1.681			
S	3/8	2.562	1.820			
T	3/8	2.750	2.030			
V	3/8	2.875	2.142			
W	3/8	3.000	2.280			
X	3/8	3.125	2.427	4-5/8 ± 1/8	2-1/8 ± 3/32	2-9/16 ± 3/32
Y	3/8	3.250	2.545			
Z	3/8	3.437	2.673			

1/ Minimum diameter of size of piece 1 before molding.

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3.4 Resistance to accelerated aging. - Packing shall withstand the test specified in 4.5.1 without hardening or undergoing any change which indicates deterioration.

3.5 Molding. - The rounded, class 1, packing shall be capable of being readily molded or prefabricated into packing sets. This capability shall be determined by the test described in 4.5.3.

3.6 Performance. - Packing shall be capable of remaining tight indefinitely, and shall have no deleterious effects upon the sheathing of electric cables or upon the metallic armor of electric cables, where used against water, air, or gas at pressures from atmospheric to 25 pounds per square inch (p.s.i.) at temperatures up to 212° F. (see 4.5.2).

3.7 Compressibility. - Packing shall be formed under the minimum pressures specified in table II with the packing and molds maintained at a minimum temperature of 100° F. When tested as specified in 4.5.3, the height shall be not less than 80 percent of the original height. For packing, class 1, furnished in bulk, spools or reels, check compressibility of all diameter of cable and packings in mold sizes equal to stuffing tube sizes C, L, R and V, respectively, in the above manner.

Table II - Molding pressures.

Type	Size	Minimum pressure (p.s.i. of annular area of packing)
I	A - J	600
I	K - Z	400
IA, II	A - R	600
IA, II	S - Z	400

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction tests. - Preproduction tests shall consist of the tests specified in 4.5 (see 6.3).

4.3 Sampling quality conformance inspection. -

4.3.1 Inspection lot. - All packing of the same class, part, type, or size shall be considered a lot for purposes of acceptance inspection.

4.3.2 Sampling for examination. - Sample sets or spools shall be selected at random in accordance with Standard MIL-STD-105 at Inspection Level I for the examination of 4.4.1. The Acceptable Quality Level shall be 2.5 percent defective.

4.3.3 Sampling for tests. - Unless otherwise specified (see 6.2), sample sets or spools shall be selected at random in accordance with Standard MIL-STD-105 at Inspection Level L4 for tests specified in 4.4.2.

4.3.4 Sampling for compression test. - Unless otherwise specified (see 6.2), a random sample of sets or spools shall be selected from each lot in accordance with 4.3.3 and subjected to the test specified in 4.5.3.

4.3.5 Sampling for production check test. - Three yards of rounded packing and two sets preformed packing from the first lot of each type and class and 1-1/2 yards of rounded packing and one set of preformed packing from every tenth lot shall be selected from the same class and type for the tests specified in 4.5. The samples shall be chosen so that over a period of time a representative selection of sizes will be sampled.

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4.4 Quality conformance inspection. -

4.4.1 Visual and dimensional examination. - Each of the samples sets or spools (see 4.3.2), shall be visually and dimensionally examined to verify compliance with this specification. Any set or spool in the sample containing one or more visual or dimensional defects shall not be offered for delivery and if the number of defective sets in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.4.2 Quality conformance tests. - Unless otherwise specified (see 6.2), quality conformance test shall include the tests specified in 4.5 for each size, type and class of packing.

4.4.2.1 Compression test. - Each of the sample sets or spools selected in accordance with 4.3.4 shall be tested in accordance with 4.5.3 to verify compliance with this specification. If any of the packing does not meet the requirements of this test, this shall be cause for rejection of the lot represented by the sample.

4.4.2.2 Production check test. - The sample selected as specified in 4.3.5 shall be subjected to the tests specified in 4.5.

4.4.2.2.1 Action in case of failure. - Lot acceptance and rejection will normally be on the basis of the sampling, examination and tests specified in 4.3, 4.4.1, and 4.4.2.1 only, and acceptance will not be withheld pending receipt of the test report on the production check test samples. However, upon receipt of an unsatisfactory report on any sample chosen for production check tests, additional samples shall be selected from every subsequent lot offered for delivery. These samples shall be subjected to the tests wherein failure was observed. Lots shall then be accepted only upon receipt of a satisfactory test report on the samples so selected. This additional testing shall be discontinued and lot acceptance returned to the normal basis when two successive lots have been accepted.

4.5 Test procedures. -

4.5.1 Accelerated aging test. - Specimens shall be suspended in an electrically heated oven and shall be subjected to dry circulating air at 200°F.

4.5.2 Performance test. - This test shall be conducted on representative sizes of bulkhead stuffing tubes. Both plain and armored cable (including sheathing material of polychloroprene, impervious resins and thermoplastic resins) shall be used during the test. After the packing has been installed in the tubes, the gland units shall be drawn up snugly with a wrench 20 inches in length. An air pressure of 25 p.s.i. for bulkhead stuffing tubes shall then be applied within the apparatus and any leakage that occurs from the tubes shall be noted and recorded. The air, with pressure still applied within the apparatus, shall be heated to a temperature of 212°F. by means of a water bath surrounding the compartments. These conditions shall be maintained for 1-1/2 hours and any leak which occurs from the tubes shall be noted and recorded. At the end of this time the apparatus shall be allowed to cool and, when cold, the leakage from the tubes shall again be recorded. The above conditions shall be repeated at least three times. On conclusion of the test, the packing shall be removed from the tubes and the cables examined for crimpage, deformation, or evidence of gross damage by chemical attach of armor or cable sheath. If any of these mentioned defects are in evidence, this shall be cause for rejection of the lot represented by the sample.

4.5.3 Compression test. - The adequacy of the molding pressures, as specified in 3.7 shall be determined as follows: Measure the height dimension (h-1, h-2, or h-1A of figures 1 and 2, as applicable)of a sample of preformed packing. Place the sample in the corresponding size mold and subject it to the pressure specified in 3.7 for a period of 3 minutes. Release pressure and measure the same dimension as soon as the set is removed from the mold.

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5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation. -

5.1.1 Basic equipment. -

5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking. - Shipment marking information shll be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made by a Government activity (see 6.2).

5.2.1.1 Packaging, packing and marking. - Items shall be packaged Level A or C as specified (see 6.2), packed Level A or B as specified (see 6.2) and marked in accordance with the requirements of Specification MIL-E-17555.)

6. NOTES

6.1 Intended use. - The packing covered by this specification is intended for use for stuffing tubes for electric cables (see table III).

Table III-Assignment of packing types, class 2

Stuffing tubes		Plugs and outlets	
BUSHIPS Drawing number	Packing type required	BUSHIPS Drawing number	Packing type required
9-S-4436-L, ALT 9 or later	1	9-S-4799-L	2
9-S-5166-L, ALT 9 or earlier	1A	9000-S6202-74438	2
9-S-5166-L, ALT 10 or later	1		

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class, part, type and size required (see 1.2, 3.7.2 and 3.3.3).
(c) If a manufacturer's respective size of packing has passed all tests on a previous submission and materials and manufacture have not been changed only tests of 4.5.3 will be required.
(d) Preservation, packaging, packing or marking requirements other than those required by paragraph 5.1 (see 5.2).

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6.3 Preproduction.-

6.3.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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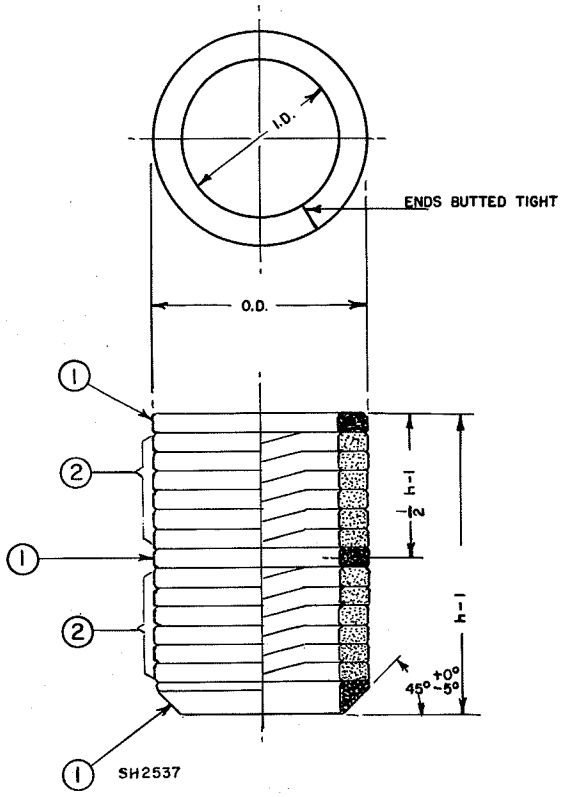


Figure 1 - Type I (sizes K to Z).

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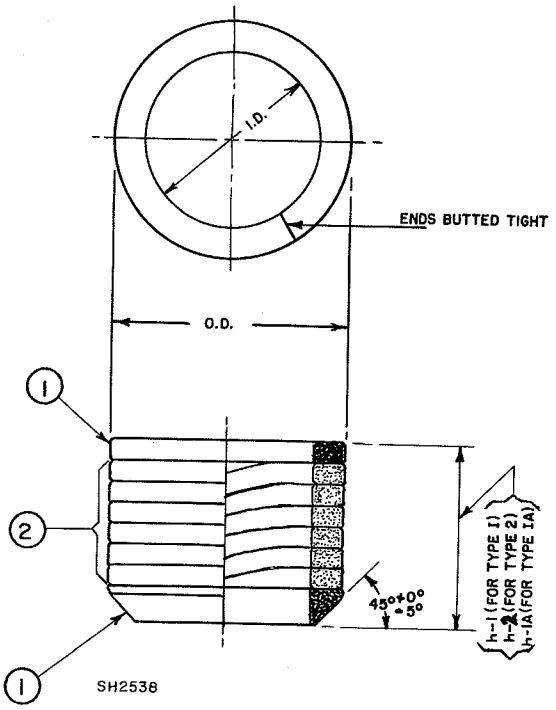


Figure 2 - Type I sizes (A to J).
Type IA (all sizes).
Type II (all sizes).

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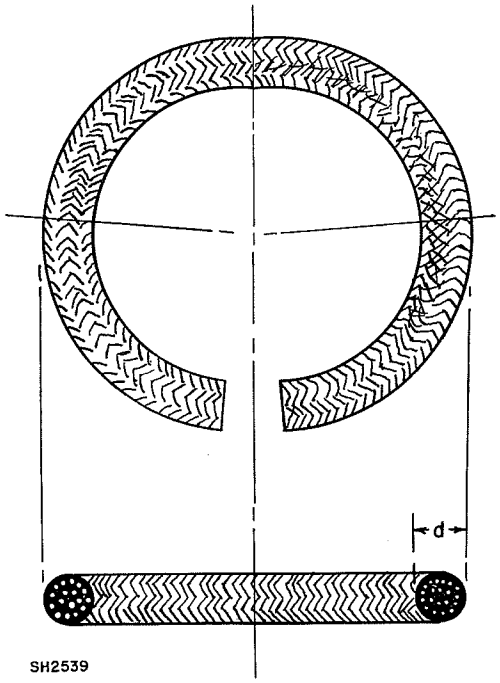


Figure 3 - Piece 1.

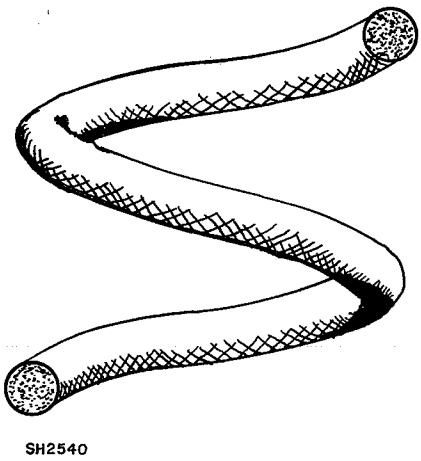


Figure 4 - Piece 2.

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

BUDGET BU. NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY *(Print name and activity)*	DATE

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